TABLE OF CONTENTS

Page

12. ENVIRONMENTAL MONITORING AND AUDIT REQUIREMENTS 12-1

12.1 Introduction 12-1
12.2 Air Quality 12-1
12.3 Noise Impact 12-1
12.4 Water Quality Impact 12-2
12.5 Waste Management Implication 12-2
12.6 Land Contamination 12-2
12.7 Ecological Impact 12-3
12.8 Fisheries 12-3
12.9 Landscape and Visual Impact 12-3
12.10 Cultural Heritage 12-4

List of Appendices

Appendix 12.1 Implementation Schedule

12. ENVIRONMENTAL MONITORING AND AUDIT REQUIREMENTS

12.1 Introduction

12.1.1 This section elaborates the requirements of environmental monitoring and audit (EM&A) for the construction and operational phases of the Project, based on the assessment results of the various environmental issues. Details of the EM&A programme and the specific monitoring requirements are presented in a stand-alone EM&A Manual.

12.1.2 The purpose of the EM&A programme is to ascertain and verify the assumptions implicit to, and accuracy of, EIA study predictions. The EM&A programme includes the scope of the EM&A requirements for the Project to ensure compliance with the EIA study recommendations, to assess the effectiveness of the recommended mitigation measures. The following sections summarise the recommended EM&A requirements with the Project Implementation Schedule presented in **Appendix 12.1**.

12.2 Air Quality

12.2.1 EM&A for potential air quality impacts should be conducted during construction phase so as to check compliance with the legislative requirements. Details of the monitoring and audit programme are contained in a stand-alone EM&A Manual.

12.2.2 Regular site audit for potential air quality impact, continuous dust monitoring is recommended to be conducted during the entire construction phase of the Project so as to ensure the air quality mitigation measures, including those stipulated in Air Pollution Control (Construction Dust) Regulation are implemented in order and to ensure that no nearby air sensitive receivers (ASRs) will be subject to adverse air quality impact.

12.2.3 No adverse air quality impact is anticipated during the operation of the Project. No EM&A would be required during the operation phase of the Project.

12.3 Noise Impact

12.3.1 Noise monitoring is recommended as part of the EM&A programme for the construction phase of the Project to check compliance with the construction noise criteria. The implementation of the recommended mitigation measures for construction activities as part of the EM&A programme, with reference to the latest implementation schedule in Construction Noise Management Plan (CNMP) to be submitted and agreed with Director of Environmental Protection (DEP).

12.3.2 There would be no adverse road traffic noise impact anticipated from the Project with implementation of the proposed mitigation measures. At-source mitigation measure (i.e. low noise road surfacing) is recommended on part of the Project roads, while at-receiver mitigation measures (e.g. acoustic windows/balconies) are recommended at some of the proposed residential developments. Road traffic noise levels should be monitored at the existing noise sensitive receivers (NSRs) located in the vicinity of the recommended at-source mitigation measures, during the first year of road opening. For the planned residential developments (Res 6), noise monitoring should be carried out during the first year of population intake. Details of EM&A programme are provided in a stand-alone EM&A Manual.

12.3.3 With proper implementation of noise control measures, with reference to Section 4, no adverse fixed noise impact would be anticipated from the proposed Pumping Station. The specification, mitigation measures, shall make reference to the latest implementation schedule as stated in the Fixed Noise Sources Management Plan (FNMP) and corresponding quantitative fixed noise assessment to be submitted and agreed with DEP. Nevertheless, as part of the design process, monitoring of operational noise from the proposed fixed plants during the testing and commissioning stage and monitoring and audit requirements would be required in the recommendations in the FNMP to verify the compliance of the EIAO-TM criteria.

12.4 Water Quality Impact

12.4.1 With full implementation of the mitigation measures such as good site practices, storm water pollution control plan, proper construction method, etc, no adverse impact is anticipated. No residual impact and cumulative impact are anticipated during both the construction and operation phases of the Project. In order to ensure effectiveness of the implemented mitigation measures, regular water quality monitoring in the LMC Meander, Ping Hang Stream and Ma Tso Lung Nullah/Stream is recommended during the construction phase.

12.4.2 No adverse water quality impacts would be expected from the operation road works if the recommended mitigation measures are implemented properly. No water quality monitoring is therefore considered necessary.

12.5 Waste Management Implication

12.5.1 The Contractor should be responsible to ensure that all waste produced during the construction of the Project are properly handled, stored and disposed of in accordance with good waste management practices, relevant legislation and waste management guidelines.

12.5.2 Waste materials generated during construction activities, such as construction and demolition (C&D) materials, are recommended to be audited at regular intervals to ensure that proper storage, transportation and disposal practices are being implemented. This would ensure the waste generated would be properly disposed of. The Contractor should be responsible for the implementation of any mitigation measures to minimise waste or mitigate problems arisen from waste materials.

12.5.3 A Waste Management Plan (WMP) should be prepared and submitted by the Contractor to the Engineer for approval. The recommended mitigation measures in this EIA Report should form the basis of the WMP. The monitoring and auditing requirement stated in ETWB TC (W) No.19/2005 should be followed with regard to the management of C&D materials.

12.5.4 Adverse environmental impacts associated with handling, storage and disposal of waste are not expected from the operation of the Project with the implementation of good waste management practices. Therefore, waste monitoring and audit programme for the operation phase of the Project would not be required.

12.6 Land Contamination

12.6.1 Remediation works, if necessary, would be carried out based on the recommended further works outlined in **Section 7.7**. Mitigation measures as recommended in the future EPD approved RAP should be implemented during the remediation works. The

EM&A requirements should be carried out in the form of regular site inspection to ensure the recommended mitigation measures are properly implemented.

12.7 Ecological Impact

12.7.1 Ecological resources were identified as requiring mitigation measures, as outlined in **Section 8**. These include wetland reinstatement and enhancement, woodland compensation, pre-construction survey with associated transplantation/translocation of species of conservation importance/ breeding and nesting behaviours, provision of temporary aboveground wildlife crossing and permanent wildlife underpass, as well as monitoring on flight path and construction time restrictions. It is recommended that these measures be monitored and audited by qualified ecologist(s) with at least 5 years of relevant experience throughout the construction phase and during the operational phase. This will ensure proper implementation and, where appropriate, monitor the performance of the proposed mitigation measures. Furthermore, a monthly site audit should be carried out throughout the construction phase to ensure that the recommended ecological mitigation measures, as well as pollution control measures, are properly implemented. In case of non-compliance, the contractor should be informed and instructed to strengthen the proposed measures accordingly. Detail of mitigation measures are provided in the standalone EM&A Manual.

12.8 Fisheries

12.8.1 The pond areas to be lost do not account for a significant proportion of the total fishpond area in Hong Kong, therefore the fisheries impact is considered to be negligible. Indirect impacts arising from construction activities and operation of the proposed development would be properly mitigated through standard practices and thus no significant fisheries impact is anticipated. Overall, no unacceptable fisheries impact is predicted from this project. As no significant fisheries impact is anticipated, monitoring of fisheries is not recommended.

12.9 Landscape and Visual Impact

12.9.1 The EIA has recommended landscape and visual mitigation measures to be undertaken during construction and operation phases of the Project. The design, implementation and maintenance of landscape and visual mitigation measures should be checked to ensure that they are fully realised and that potential conflicts between the proposed landscape measures and any other project works and operation requirements are resolved at the earliest possible date and without compromise to the intention of the mitigation.

12.9.2 The construction phase EM&A of the mitigation measures shall be carried out as part of the site audit programme. EM&A during operation phase of the Project shall be carried out within the 12-month establishment period of the landscape and visual mitigation measures by the corresponding implementation agency to ensure the proposed mitigation measures in the EIA and as depicted in the Landscape and Visual Mitigation Plan are fully implemented.

12.9.3 All mitigation measures proposed in the EIA and implemented by the Contractor should be audited by Registered Landscape Architect (RLA), as a member of the Environmental Team, on a regular basis to ensure compliance with the intended aims of the measures. The mitigation measures proposed should be embodied into the detailed engineering design and landscape design drawings and contract document. Site inspection should be undertaken monthly throughout the construction period. In

particular, the extent of the agreed works areas should be regularly checked during the construction phase. The landscape auditor should audit the proposed mitigation measures in the EIA to ensure that they are fully implemented during construction and the 12-month establishment period during operation phase.

12.10 Cultural Heritage

12.10.1 MacIntosh Fort (Ma Tso Lung) (HB303) (grade 2 historic building) is situated within 50m from the Site Boundary. Potential indirect impacts or ground borne vibration, tilting and settlement would be anticipated during the construction phase. Therefore, during construction phase, monitoring of ground-borne vibration, tilting and ground settlement is proposed to be employed. The monitoring for HB303 should be incorporated with a set of Alert, Alarm and Action (3As) system following AMO's monitoring requirements for grade 2 historic building, as well as vibration-sensitive and dilapidated buildings against continuous vibration. The values for building settlement markers are clarified for monitoring both the building uplift and settlement in the vertical direction. Furthermore, pre and post condition survey in construction phase is required. An Authorized Person (AP), as well as relevant professionals, such as Registered Structural Engineer (RSE), should be employed to carry out the condition survey and structural assessment to inspect the physical condition and structural integrity of the building and provide advice on the subsequent mitigation measures based on the assessment result. A detailed monitoring proposal is required to be submitted for AMO's agreement prior to the commencement of the proposed works including (a) actual checkpoint locations, (b) installation details, (c) response actions for each of Alert/ Alarm/ Action (3As) levels, (d) monitoring frequency and (e) mitigation measures. Regular and interval inspections or condition surveys may be considered if any irregularities are observed from the monitoring record, with circulation to AMO for record. Details of the proposed construction works within 50m distance of the above-mentioned, along with the impact assessment and mitigation measures when necessary, shall be submitted for AMO's consideration at the detailed design stage.

12.10.2 No site of archaeological interest (SAI) is identified within assessment area. Meanwhile, the Site Boundary and its 300m assessment area is of no archaeological potential. Therefore, no archaeological impact would be anticipated. Thus, no mitigation measure would be required. As a precautionary measure, AMO should be informed immediately in case of discovery of antiquities or supposed antiquities in the course of works, so that appropriate mitigation measures, if needed, can be timely formulated and implemented in agreement with AMO.